

JavaScript Activity: Comparisons, Booleans, and Template Literals

Learning Goals

By the end of this activity, students will be able to:

- Understand Boolean values (true / false)
- Use relational comparison operators (> < >= <= !=)
- Understand the difference between loose equality (==) and strict equality (===)
- Use template literals with `${ }` to insert variables into strings
- Format console output using multi-line template literals

Step 1 – Create a JavaScript File

Create a file named `testScores.js` and add the following starter code:

```
"use strict";
```

```
let test1 = 95;
```

```
let test2 = 56;
```

Step 2 – Compare the Scores

Create variables that compare the two test scores.

```
let compareTest1 = test1 > test2;
```

```
let compareTest2 = test1 < test2;
```

Step 3 – Display Results Using Concatenation

Print the results using string concatenation.

```
console.log("Is test 1 greater than test 2? " + compareTest1);
```

```
console.log("Is test 1 less than test 2? " + compareTest2);
```

Step 4 – Display Results Using Template Literals

Rewrite the same statements using template literals. Template literals use backticks (```) and allow variables inside `${ }`.

```
console.log(`Is test 1 greater than test 2? ${compareTest1}`);
```

```
console.log(`Is test 1 less than test 2? ${compareTest2}`);
```

Step 5 – Try More Comparison Operators

```
compareTest1 = test1 >= test2;
```

```
compareTest2 = test1 <= test2;
```

```
console.log(`Is test 1 greater than or equal to test 2? ${compareTest1}`);
```

```
console.log(`Is test 1 less than or equal to test 2? ${compareTest2}`);
```

Step 6 – Compare Words (Case Sensitivity)

```
let done = "done";  
let DONE = "DONE";
```

```
let caseSensitive = done == DONE;
```

```
console.log(`Is "done" the same as "DONE"? ${caseSensitive}`);
```

JavaScript is case-sensitive, so these values are considered different.

Step 7 – Loose Equality (==)

Loose equality compares values but allows type conversion.

```
console.log(`Is 5 equal to "5"? ${5 == "5"}`);
```

Step 8 – Strict Equality (===)

Strict equality compares BOTH the value and the data type.

```
console.log(`Is 5 strictly equal to "5"? ${5 === "5"}`);
```

Step 9 – Grade Comparison Challenge

```
let englishGrade = 87;  
let scienceGrade = 76;
```

```
console.log(  
  ----- Grade Comparison -----
```

```
  English Grade: ${englishGrade}  
  Science Grade: ${scienceGrade}
```

```
  Is English greater than Science? ${englishGrade > scienceGrade}
```

```
  Are the grades equal using == ? ${englishGrade == scienceGrade}
```

```
  Are the grades strictly equal using === ? ${englishGrade === scienceGrade}
```

```
  -----  
`);
```

Student Challenge

Create two new variables and compare them.

```
let scoreA = 90;  
let scoreB = "90";
```

Use template literals to answer:

1. Are the scores equal using `==` ?
2. Are the scores equal using `===` ?
3. Is `scoreA` greater than `scoreB`?

Reflection Questions

1. What symbol is used to create template literals?
2. What does `${ }` do inside a template literal?
3. What is the difference between `==` and `===` ?